The Pareto Landscape of Moral Reasoning

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Abstract

Evaluating moral reasoning is hard because the moral domain lacks the ground truth that organises evaluation elsewhere: competent reasoners can start from similar intuitions and reach different defensible conclusions. We argue that this is not a measurement failure but a structural feature, and we propose a conceptual framework that takes it seriously. Moral reasoning, on our account, aims at a *good epistemic state*: a pair of commitments and principles that hangs together. Such a state is graded along two axis families that genuinely conflict: epistemic desiderata (account, systematicity, faithfulness) and value loadings (freedom, well-being, justice, dignity, solidarity). The rational object of study is then the *Pareto frontier* of attainable epistemic states. We develop a vocabulary for the shapes this frontier takes (convex segments, concave segments, knees, plateaus, corner attractors, dominated regions), and we recast familiar reasoning methods, reflective equilibrium foremost, as strategies for navigating the landscape toward the frontier. We sketch the corpus that would let this framework be studied empirically, and outline what such study would reveal about moral reasoning in humans and in language models.

1 Motivation

Current alignment work focuses on the moral *content* a model holds: constitutional principles, character training, value specifications. Less attention has gone to moral reasoning as a *capacity* (Carlsmith, 2026), and existing evaluations tend to conflate the two (Haas et al., 2026). But a value-aligned model whose reasoning fails when its values come into tension is still unsafe: it can produce outputs that look like principled moral thought without being so. Diagnosing such failures is hard because the moral domain lacks a ground truth against which to score answers (Queloz, 2025): competent reasoners can reach different conclusions without either being simply wrong.

This paper proposes a framework that takes the absence of ground truth as a *structural feature* rather than a measurement failure. We articulate what moral reasoning aims at, what makes its outputs better or worse without a single right answer, and how the multi-objective optimisation literature (Deb, 2001) gives us a precise vocabulary for the kinds of trade-offs the moral domain actually exhibits. Familiar methods such as reflective equilibrium (Rawls, 1971; Knight, 2023), recently formalised by Beisbart et al. (2021) and studied at simulation scale by Freivogel (2023), reappear as *strategies for navigating* this landscape, comparable in principle to alternative strategies and amenable to empirical study once a suitable corpus is in place.

2 Conceptualising moral reasoning

An **ethical situation** confronts an agent with a decision: several actions are open, and none is clearly right or wrong. On encountering such a situation the agent has **commitments**: pre-theoretical verdicts about how to act in the case, of the form “in situation s, action a is required / permissible / forbidden.”

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Across the body of situations the agent has considered, these commitments form a set C. A reasoner holding C alone has no resources to handle a new situation, to defend a verdict against an objection, or to notice when two commitments pull in opposite directions.

This is where **principles** enter. A principle p is a general moral rule, independent of any particular situation, that an agent endorses as justifying some of their commitments. The agent's principles together form a set P. Principles do three kinds of work simultaneously: they *account* for commitments (giving each commitment a reason), they *generalise* to situations the agent has not yet considered (a single principle yields verdicts across many cases), and they *regulate* the commitments (revealing inconsistency when two commitments cannot both follow from the same principle).

The output of moral reasoning is an **epistemic state** $\sigma = (C, P)$: a set of commitments paired with a set of principles that hangs together. We take the goal of moral reasoning to be reaching a *good* epistemic state. Two families of demand pull on what *good* means here. The state should satisfy the agent's *epistemic* expectations of reasoning: the principles should account for the commitments, they should be simple and broadly applicable rather than a hodgepodge of one-offs, and the commitments should not drift arbitrarily from the agent's starting intuitions C_0. The state should also fit the agent's *ethical* profile: the principles should express the values and deeper ethical frameworks the agent endorses on reflection, and where values trade off, the trade-off the state strikes should be one the agent can stand behind. We make these two families precise in Section 3.

Reasoning moves as typed operations. Reaching a good σ requires recognisable operations on C and P. We list the primitives with their input-output types, which makes each addressable as a unit of evaluation independent of the strategy that composes it.

- **Property elicitation** $s \to \{\varphi_1, \ldots, \varphi_m\}$. From a situation, make explicit the ethically relevant properties: who is affected, what interests are at stake, the kind of action on offer. This is the basis on which a commitment will be formed.
- **Principle proposal** (abduction) $\{c_1, \ldots, c_n\} \to p$. From a set of commitments, propose a candidate principle that justifies them. The inductive move from cases to rule.
- **Principle deduction** $(p, s) \to c$. From a candidate principle and a situation, derive the commitment the principle implies. The downward complement of proposal.
- **Commitment revision** $(c, p) \to c'$. Adjust a commitment in light of a principle that conflicts with it.
- **Principle revision** $(p, c) \to p'$. Adjust a principle in light of a commitment that conflicts with it.
- **Coherence checking** $\sigma \to \mathbb{R}^{3+K}$. Score the current state on the epistemic and ethical axes (Section 3). A meta-operation: it does not move σ, but it tells any other operation whether its candidate move improves things.

The dialectical alternation of the two revisions is the move at the heart of reflective equilibrium (Rawls, 1971; Beisbart et al., 2021), and it is what gives that strategy its name. We make the strategy explicit in Section 5, after spelling out the space the moves take place in.

3 What makes an epistemic state good

Two axis families grade σ simultaneously. They genuinely conflict, in the sense that improving one typically costs another.

Epistemic axes. Properties of σ as reasoning, drawn from the formal account of Beisbart et al. (2021).

- ACCOUNT: how much the principles in P entail or support the commitments in C, penalising contradictions, gaps, and surplus content.
- SYSTEMATICITY: how simple, unifying, and broadly applicable P is, as opposed to a hodgepodge of case-by-case stipulations.
- FAITHFULNESS: how well the current commitments respect the agent's initial intuitions C_0, penalising arbitrary drift.

Ethical axes. Properties of σ as a moral stance. Each principle $p \in P$ carries a fuzzy *value loading* $\boldsymbol{v}(p) \in [0,1]^K$ over a small set of basic moral values. We work with five: FREEDOM, WELL-BEING, JUSTICE, DIGNITY, and SOLIDARITY, chosen so that the central tensions of moral life (liberty vs. community, fairness vs. welfare, welfare vs. dignity) appear directly as trade-offs among axes. The state σ inherits an aggregate value position by aggregating its principles' loadings: the more a value is activated by the principles in P, the more the state weighs on that axis. Concretely, $V(\sigma) = \frac{1}{|P|}\sum_{p \in P} \boldsymbol{v}(p) \in [0,1]^K$, so locating an epistemic state in the ethical-value subspace amounts to checking which values each of its principles activates.

Two families, two axes of disagreement. The epistemic family is broadly agent-independent: every reasoner has some stake in coherence. The ethical family is agent-dependent: different agents weight different values, and this pluralism is a constitutive feature of the moral domain rather than noise to be averaged out.

Pluralism is not subjectivism. Although the choice of a point on the frontier is agent-dependent, the frontier itself is not: the set of defensible epistemic states can in principle be characterised independently of any value profile. This suggests a productive division of labour for an ideal artificial moral reasoner: map the full frontier and present its structure, leaving the deliberative choice of point to human reasoners. Figure 1 illustrates the two-subspace structure: the same state σ has coordinates in both the epistemic subspace Π_E and the ethical-value subspace Π_V.

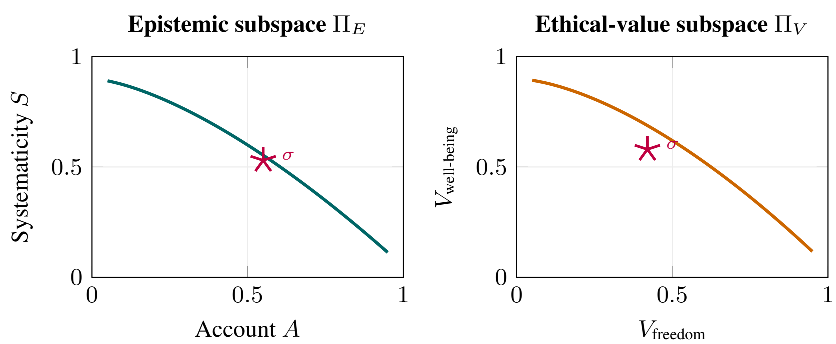

Figure 1: **Two subspaces, one epistemic state (schematic).** A state σ has coordinates in the epistemic subspace (left) and the ethical-value subspace (right). The two projected frontiers Π_E and Π_V need not coincide; a good σ lies on both.

4 The space in which moral reasoning evolves

We can now state the framework precisely. Let Σ be the set of all internally consistent epistemic states. Each $\sigma \in \Sigma$ has a position in the joint objective space

$$\big(A(\sigma),\, S(\sigma),\, F(\sigma \mid C_0),\, V_1(\sigma), \ldots, V_K(\sigma)\big) \in [0,1]^{3+K}.$$

Multi-objective optimisation (Deb, 2001) gives us the right vocabulary for the structure of this space when its axes genuinely conflict.

Pareto dominance. A state σ' *Pareto dominates* σ, written $\sigma \prec \sigma'$, if every objective at σ' is at least as large as at σ and at least one is strictly larger.

The Pareto frontier. The frontier Π is the set of non-dominated states in Σ:

$$\Pi \;=\; \{\, \sigma \in \Sigma : \nexists\, \sigma' \in \Sigma \text{ with } \sigma \prec \sigma' \,\}.$$

Π is what we mean by *the space of defensible epistemic states*. For any state off the frontier, some defensible move strictly improves it on every axis the agent cares about; for any state on the frontier, no such move exists.

Figure 2 sketches the picture at the heart of the paper. The Pareto frontier collects the defensible epistemic states for a given case; reasoning strategies are trajectories through the space, and different strategies can land at different points on the frontier (or fail to reach it). Reflective equilibrium is one trajectory among several.

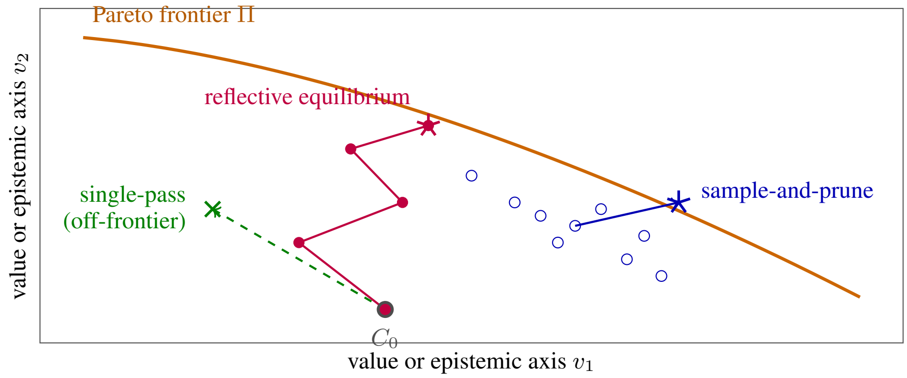

Figure 2: **Reasoning as movement toward the Pareto frontier (schematic).** The frontier Π collects the defensible epistemic states. Different reasoning strategies trace different paths through the space. Reflective equilibrium walks dialectically from the initial commitments C_0 and ends at a frontier point. Sample-and-prune explores broadly and selects the non-dominated candidate. Single-pass abduction leaps once and may land in the dominated region. Shapes and trajectories are stylised, not derived from data.

Reasoning traces. A reasoning trace is a finite sequence $\sigma_0, \sigma_1, \ldots, \sigma_n$ of epistemic states linked by moves of the kinds enumerated in Section 2. The trace has a *shape* in the joint space, and each move has a characteristic contribution. Principle proposal typically lifts ACCOUNT and SYSTEMATICITY together. Commitment revision trades FAITHFULNESS for ACCOUNT. Principle revision trades SYSTEMATICITY for ACCOUNT, or one value for another.

Vocabulary for the shape of the frontier. Different ethical situations, and sets of situations grouped into a context or domain, produce Pareto frontiers of different shapes. The multi-objective optimisation literature already has a rich vocabulary for the shapes these frontiers can take, which we can borrow directly.

- **Convex segments.** Weighted sums of axes pick a unique answer. There is little genuine disagreement: reasoners differ in weights but agree in conclusion.
- **Concave segments.** Incomparable defensible answers: no weighting of axes selects between them. *This is where genuine pluralism lives.*
- **Knee points.** A small concession on one axis buys a large gain on another. The *natural compromise* of the case, where reasoning tends to land when neither value dominates.
- **Plateaus.** Many near-equivalent positions. The choice is under-determined and competent reasoners may land in slightly different places without error.
- **Corner attractors.** Extreme positions where one axis dominates and others collapse. Pure utility, pure autonomy, pure procedural justice. Often degenerate.
- **Dominated region.** Bad reasoning. Off-frontier states are off-frontier precisely because some defensible move strictly improves them.

Figure 3 illustrates each shape on a stylised two-axis front.

An illustrative example. Consider a user who refuses a prescribed medication, partly because of delusional thinking, while interacting with an LLM acting as a therapeutic interlocutor. The values primarily in tension are FREEDOM and WELL-BEING. Two responses sit on a concave segment of Π: an autonomy-leaning one states the medical facts plainly and does not push further, and a care-leaning one explores the refusal through dialogue and non-manipulative persuasion. Neither dominates the other. An interior knee point may exist (a specific autonomy-preserving care strategy that buys large WELL-BEING gain for small FREEDOM cost). Several responses are off-frontier and

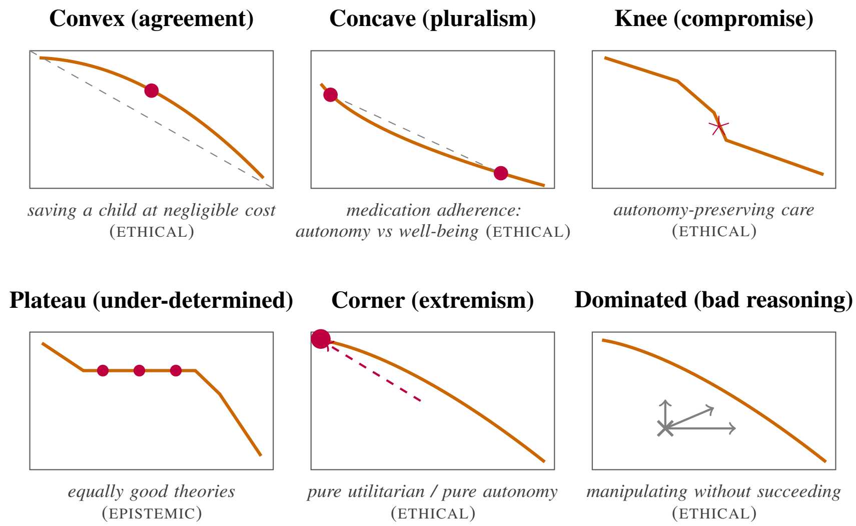

Figure 3: **Shapes the Pareto frontier can take (schematic).** Each panel illustrates one shape concept with a sample ethical or epistemic context where one would expect to find it. Stylised, not derived from data.

dominated: insulting the user, manipulating without succeeding, walking away without escalation. Figure 4 visualises this landscape.

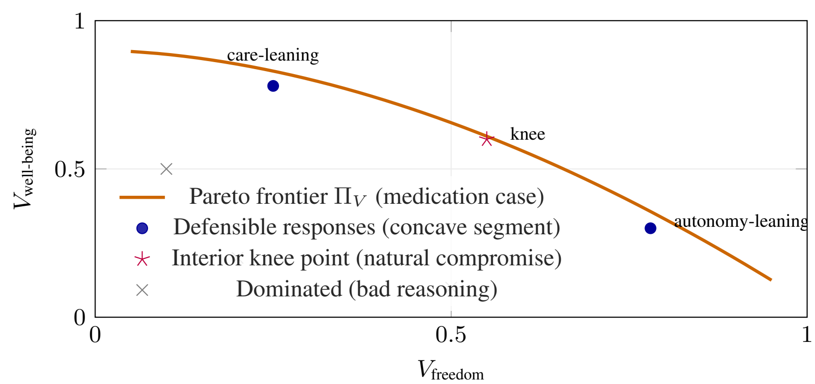

Figure 4: Sketch of the ethical-value Pareto frontier in the medication case, projected onto FREEDOM and WELL-BEING. Two defensibly different responses sit on a concave segment; an interior knee point captures a natural compromise; several reachable responses are strictly dominated.

5 Strategies for navigating the space

If moral reasoning aims at the frontier, the natural follow-up question is *how an agent gets there*. Reflective equilibrium is the philosopher-favoured answer, but it is one strategy among several. We sketch the space of strategies and the primitives they share.

Shared primitives. Any strategy that walks toward Π composes the typed operations introduced in Section 2: principle proposal, principle deduction, commitment revision, principle revision, and coherence checking. Strategies differ in *how* they sequence and compose these primitives, but the primitives themselves are the capacities an artificial moral reasoner must possess. For evaluating such a reasoner, the primitives are the natural unit of analysis.

Reflective equilibrium. Beginning from the initial commitments C_0 and an arbitrary P, alternate revisions of P and C, each step guided by current scores (Beisbart et al., 2021; Freivogel, 2023). The trace is local: each move depends on the current state, and the trajectory walks toward a fixed point. Reflective equilibrium is well-suited to finite reasoners with strong intuitions and limited search budget; it is the canonical strategy in the philosophical literature (Knight, 2023).

Sample-and-prune. Generate many candidate epistemic states by combining principles and commitments, score each, retain the non-dominated ones. A direct attempt to map a region of Π rather than walk toward a single point. Well-suited to reasoners with cheap sampling, perhaps less well-suited to finite human deliberation.

Tree search. Build a tree of reasoning moves from σ_0, expand promising branches, backtrack from dead ends. A natural fit for artificial reasoners with a notion of branch quality and a backtracking mechanism.

Single-pass abduction. Propose one principle from C_0, deduce its consequences, accept the result. The naive baseline. Often under-evaluated against the others.

Multi-start equilibrium. Run reflective equilibrium from several distinct starting points to discover multiple regions of Π at once. Particularly useful when the frontier is expected to be concave or plateau-shaped.

Corner-collapse strategies. Maximise a single axis (pure ACCOUNT, pure WELFARE, pure AUTONOMY). Degenerate in general but instructive as ablations and as failure modes of stronger strategies.

Which strategy for which reasoner? For humans, reflective equilibrium is the canonical answer, in part because human deliberation has a budget that favours local moves over wide sampling. For artificial reasoners (LLMs in particular), the question is open. Sampling-based strategies may exploit parallelism that humans cannot; tree-search strategies may map richer landscapes; the failure modes of single-pass strategies may or may not appear under chain-of-thought. The framework lets us ask these questions on equal footing once we can locate any strategy's output in the space.

6 A corpus to make this empirical

The framework so far is conceptual; turning it empirical requires a corpus. We sketch what we would dream of having and what it would let us do.

Contents. For a body of ethical situations, the corpus would contain three layers.

1. **Commitment pool** $\mathcal{C}$: candidate verdicts on each situation. Per-case action judgments of the form "in situation s, action a is required / permissible / forbidden," generated broadly enough to cover the defensible verdicts plus a margin of bad ones.
2. **Principle pool** $\mathcal{P}$: candidate moral principles, each tagged with a *fuzzy value loading* $\boldsymbol{v}(p) \in [0,1]^K$ indicating to what degree the principle expresses each value. Optionally, a secondary tagging by ethical framework $\boldsymbol{f}(p) \in [0,1]^M$ (utilitarian, Kantian, virtue, care, contractualist) supports framework-level analyses without committing to frameworks as primary axes.
3. **Dialectical graph** ρ: for every principle-commitment pair, a *fuzzy inferential strength* $\rho(p,c) \in [-1,1]$, where positive values indicate degree of entailment, negative values degree of contradiction, and zero silence. This generalises a discrete entails / contradicts / silent labelling and captures the partial support relations that are typical in moral reasoning.

Construction. The corpus would be generated by a large language model and verified by philosophers at every stage: generation, fusion of near-duplicates, deletion of weak candidates, scoring of independent quality dimensions, value tagging, and inferential strengths. We treat the corpus as a reusable resource in its own right, independent of any downstream reasoning method that operates on it.

What it would let us do. Given a starting set of intuitions C_0 drawn from $\mathcal{C}$, the corpus lets us:

- Compute the Pareto fronts Π_E and Π_V of attainable epistemic states, and characterise their shapes locally and globally.
- Locate any epistemic state σ in the joint space, and say whether it is on Π, near Π, or dominated, and if dominated, by what.
- Inspect reasoning traces move by move: is each step principled (a justified abduction, a defensible commitment revision) or unprincipled (a faithfulness violation with no compensating gain)? Does the trajectory walk toward Π or away from it?
- Compare reasoning strategies on equal footing.

7 What this lets us study

The framework, combined with such a corpus, opens several lines of empirical work that are difficult to pose otherwise.

The geometry of moral disagreement. For a given case, is the local shape of Π convex (resolvable disagreement) or concave (genuine pluralism)? Does the case admit a knee point (a natural compromise)? Across many cases, what shapes recur, and what does that tell us about the structure of the moral domain?

Strategy comparison. Do different reasoning strategies reach different regions of Π on the same case? Does reflective equilibrium land in plateaus while sample-and-prune maps concave segments? Which strategy is more robust under reframing of the case?

LLM moral reasoning. A value-aligned model whose traces sit off Π in characteristic ways is a value-aligned model that reasons badly, and the framework lets us name how. Specific failure modes become diagnosable: *corner collapse* (the trace ends at an axis extreme), *anti-revision rigidity* (no move revises the principle side), *faithfulness violations* (commitments dropped with no justifying gain), *stability collapse* (different landings under semantically equivalent rephrasings of the situation). These are empirically testable hypotheses about LLM behaviour, but they only become testable once the framework names them.

Evaluating typed primitives in isolation. If the typed operations of Section 2 are the irreducible capacities of moral reasoning, then a natural question for any artificial moral reasoner is: which of those capacities does it have, and at what quality? The corpus lets us isolate each primitive and evaluate it independently of the strategy it is embedded in. We can probe, separately, how well a model forms initial commitments from raw situations, proposes principles from a held-out set of commitments, deduces verdicts from a given principle, revises commitments under a conflicting principle, revises principles under a conflicting commitment, and scores epistemic states. Each of these yields a focused diagnostic, in addition to the trace-level analyses the corpus already supports.

8 Discussion

Related work. Our framework draws on three strands that have not previously been brought together. The first is the formal model of reflective equilibrium developed by Beisbart et al. (2021) and the simulation study of Freivogel (2023), which we adopt for the epistemic desiderata. The second is the literature on moral pluralism and the asystematicity of normative domains (Queloz, 2025; Elgin, 1996), which motivates treating the absence of ground truth as structural. The third is multi-objective optimisation (Deb, 2001), from which we borrow the vocabulary of dominance, frontier, and shape. The empirical agenda our framework supports is most closely related to recent work on evaluating moral reasoning in language models (Haas et al., 2026; Awad et al., 2022), with the difference that our unit of analysis is the *trace* and its location in a measurable space rather than aggregate behaviour on a benchmark.

Outlook. The framework gives the familiar pieces of moral reasoning a clearer place: the goal is a good epistemic state, goodness is graded along epistemic and ethical axes that genuinely conflict, the rational object of study is the Pareto frontier of attainable states, and methods such as reflective equilibrium are strategies for navigating it. What remains is the corpus (Section 6) and the empirical study it supports (Section 7). Both are non-trivial to build, but both are now well-posed: the framework tells us what to measure, what shapes to look for, and what failure modes to name.

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